

1. During a 10-year period of constant inflation, the value of a \$600,000 property will increase according to the equation $v = 600,000e^{0.04t}$.
- What will be the value of the property in 9 years?
 - Use a table or graph to estimate when this property will be worth double its original value.

a. The value of the property in 9 years will be \$.
(Round to the nearest cent as needed.)

b. This property will be worth double its original value in years.
(Round to the nearest year as needed.)

- *2. Suppose the quantity (in grams) of a radioactive substance present at time t is $Q(t) = 1296(3^{-0.5t})$, where t is measured in months.

- How much will be present in 6 months?
- How long will it take to reduce the substance to 16 grams?

a. There will be grams of the substance present in 6 months.
(Round to the nearest gram as needed.)

b. It will take months to reduce the substance to 16 grams.
(Round to the nearest month as needed.)

3. Solve the logarithmic equation.

$$9 + 3 \ln x = 21$$

$x =$
(Simplify your answer. Type an exact answer, using e as needed.)

4. Solve.

$$6^{-x+3} = 61$$

$x \approx$
(Simplify your answer. Type an integer or decimal rounded to four decimal places as needed.)

5. A minute ad during a campaign in 1973 cost \$200,000. The price tag for a 30-second ad slot during a 2005 campaign was \$3 million. The cost of a 30-second slot of advertising for a campaign can be modeled by $y = 0.0000963(1.104^x)$, where x is the number of years after 1900 and y is the cost in millions.

Complete (a) and (b) below.

- a. According to the model, what was the cost of a 30-second campaign ad in 2001?

The cost of a 30-second campaign ad in 2001 was \$.
(Simplify your answer. Round to the nearest dollar as needed.)

- b. The cost of a 30-second campaign ad in 2015 was \$8.0 million. Does the model underestimate or overestimate the cost of these ads?

The model (1) _____ the cost for a 30-second campaign ad. The model estimates the cost for 2015 at \$.
(Simplify your answer. Round to the nearest dollar as needed.)

- (1) ☐ overestimates
☐ underestimates

6. If \$13,000 is invested at 6% interest compounded monthly, find the interest earned in 16 years.

The interest earned in 16 years is \$.
(Do not round until the final answer. Then round to two decimal places as needed.)

7. If \$4000 is invested at 5% compounded continuously, what is the amount after 4 years?

The amount after 4 years will be \$.
(Round to the nearest cent.)

8. What lump sum must be invested at 10%, compounded monthly, for the investment to grow to \$50,000 in 13 years?

The lump sum \$ invested at 10%, compounded monthly, grows to \$50,000 in 13 years.
(Do not round until the final answer. Then round to the nearest cent as needed.)

9. At the end of t years, the future value of an investment of \$10,000 in an account that pays 9% APR compounded monthly is

$S = 10,000 \left(1 + \frac{0.09}{12} \right)^{12t}$ dollars. Assuming no withdrawals or additional deposits, how long will it take for the investment to amount to \$40,000?

The investment will grow to \$40,000 in years.
(Do not round until the final answer. Then round to two decimal places as needed.)

10. Suppose that \$18,000 is invested at continuous rate r and grows to \$81,108 in 28 years. Find the interest rate r .

The interest rate is %.
(Round to the nearest percent as needed.)

11. What rate of interest compounded annually is required to triple an investment in 22 years?

The rate of interest required is %.
(Round to two decimal places as needed.)

- *12. Following the birth of a child, a parent wants to make an initial investment P_0 that will grow to \$40,000 for the child's education at age 19. Interest is compounded continuously at 6%.
What should the initial investment be? Such an amount is called the present value of \$40,000 due 19 years from now.

The present value is about \$.
(Do not round until the final answer. Then round to two decimal places as needed.)

13. The spread of a highly contagious virus in a high school can be described by the logistic function

$$y = \frac{5600}{1 + 699e^{-0.7x}}$$

where x is the number of days after the virus is identified in the school and y is the total number of people who are infected by the virus.

(a) What is the upper limit of the number infected by the virus during this period?

(b) How much time has gone by when 4100 people are infected?

(Round to three decimal places)

(c) How many people are infected when the disease is first diagnosed?

(d) If you round the number of infect people to the nearest whole number, how long will it take for the maximum number of people to be infected **(Round the number of days up to a whole number)**

(a) The upper limit of the number infected by the virus during the period is .

(b) The number of days that have passed by when 4100 people are infected is days

(Round to three decimal places)

(c) On the day of first diagnosis there are already infections

(d) Rounding to the nearest whole number, it will take days for the maximum number of infections to be reached **(Round the number of days up to a whole number)**